



RIPHAH
INTERNATIONAL UNIVERSITY

COMPUTER ARCHITECTURE

ASSIGNMENT #2



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BSCS

Section A: Pipelining & Space-Time Diagrams

1. [Numerical – 5 Marks]

A 5-stage instruction pipeline (IF, ID, EX, MEM, WB) executes a total of 10 instructions.

- Each stage takes **1 clock cycle**.
- Assume there are **no pipeline hazards**.

Tasks:

- Draw the **space-time (space-time diagram)** showing the execution of all 10 instructions.
- Calculate the **total time (in clock cycles)** required to complete all instructions.
- Compare it with the **non-pipelined execution time**, assuming each instruction takes 5 cycles.
- Compute the **speedup** gained by pipelining.

2. [Numerical – 5 Marks]

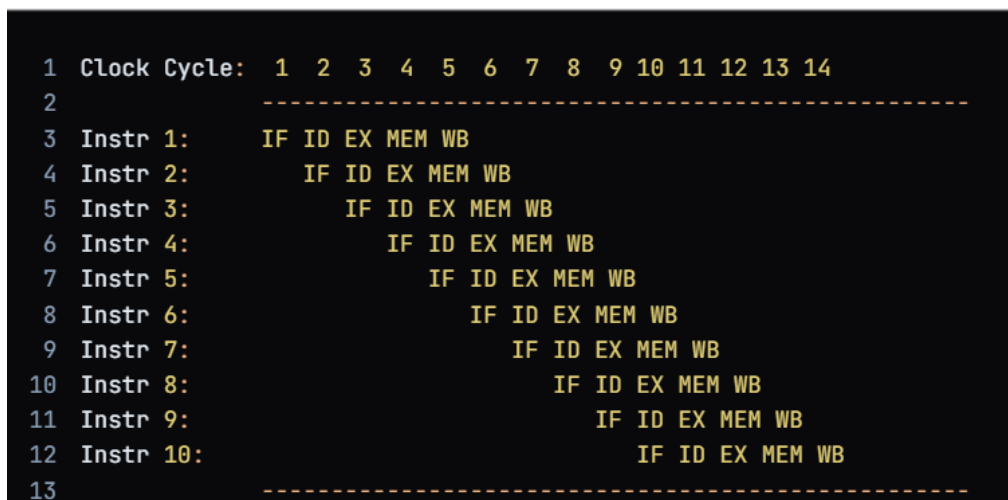
A pipeline has **4 stages**, each taking **2 ns** to complete. Due to data hazards, a **stall of 1 cycle** occurs after every 4 instructions.

- Find the **average instruction throughput** (instructions per nanosecond) for 20 instructions.
- Draw a **simplified timing diagram** indicating the stall positions.

Solution:

5-Stage Pipeline Execution (No Hazards)

a) Space-Time Diagram (built it on vs code)



1	Clock Cycle:	1	2	3	4	5	6	7	8	9	10	11	12	13	14
2		-----													
3	Instr 1:	IF	ID	EX	MEM	WB									
4	Instr 2:		IF	ID	EX	MEM	WB								
5	Instr 3:			IF	ID	EX	MEM	WB							
6	Instr 4:				IF	ID	EX	MEM	WB						
7	Instr 5:					IF	ID	EX	MEM	WB					
8	Instr 6:						IF	ID	EX	MEM	WB				
9	Instr 7:							IF	ID	EX	MEM	WB			
10	Instr 8:								IF	ID	EX	MEM	WB		
11	Instr 9:									IF	ID	EX	MEM	WB	
12	Instr 10:										IF	ID	EX	MEM	WB
13		-----													

b) Total Time Required

Total clock cycles = 5 + (10 - 1) = 14 cycles

I n s t r.	1	2	3	4	5	6	7	8	9	1 0	1 1	1 2	1 3	1 4	1 5	1 6	1 7	1 8	1 9	2 0	2 1	2 2	2 3	2 4	2 5	2 6	2 7	2 8
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[illegible]

[illegible]

[illegible]

Section B: IEEE 754 Floating Point Standard

3. [Numerical – 5 Marks]

Represent the following decimal numbers in **IEEE 754 single-precision (32-bit)** floating-point format:

- a) +13.25
- b) -0.75

Show all steps clearly:

- Convert to binary
- Normalize the number
- Determine sign bit, exponent, and mantissa
- Write the final 32-bit binary representation

4. [Numerical – 5 Marks]

Convert the following **IEEE 754 single-precision number** into its decimal equivalent:

01000001101110000000000000000000

- Identify the **sign**, **exponent**, and **mantissa**
- Show the step-by-step conversion to the **final decimal value**

Question 3: Decimal to IEEE 754 Single-Precision

a) +13.25

1. Convert to Binary:

Integer (13): $13 = 8 + 4 + 1 \rightarrow 1101$

Fraction (0.25): $0.25 \times 2 = 0.5$; $0.5 \times 2 = 1.0 \rightarrow .01$

Result: 1101.01

2. Normalize:

Shift decimal 3 places left: 1.10101×2^3

3. Determine Fields:

Sign Bit: Positive $\rightarrow 0$

Biased Exponent: $3 + 127 = 130$. Binary (8-bit): 10000010

Mantissa (23-bit): 10101000000000000000000

Final 32-bit Binary:

0 10000010 10101000000000000000000

b) -0.75

1. Convert to Binary:

Integer (0): 0

Fraction (0.75): $0.75 \times 2 = 1.5$; $0.5 \times 2 = 1.0 \rightarrow .11$

Result: 0.11

2. Normalize:

Shift decimal 1 place right: 1.1×2^{-1}

3. Determine Fields:

Sign Bit: Negative $\rightarrow 1$

Biased Exponent: $-1 + 127 = 126$. Binary (8-bit): 01111110

Mantissa (23-bit): 10000000000000000000000

Final 32-bit Binary:

1 01111110 100000000000000000000000

Question 4: IEEE 754 to Decimal Conversion

Input: 0 10000011 011100000000000000000000

1. Identify Fields:

Sign (S): 0 (Positive)

Biased Exponent (E): 10000011

Mantissa (M): 011100000000000000000000

2. Calculate Actual Exponent:

Decimal value of 10000011 = 131

Actual Exponent = $131 - 127 = 4$

3. Construct Normalized Binary:

Value = $+(1.0111)_2 \times 2^4$

4. Final Decimal Value:

Shift decimal point 4 places right: 10111.0

$$10111_2 = 16 + 4 + 2 + 1 = 23.0$$

Result: 23.0